

Abstract No.

Convergent Physics-Guided and Deep Learning Analyses Reveal a Common Prognostic Motility Axis in Glioblastoma

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Background: Cell motility drives glioblastoma (GBM) invasion, but biomarkers linking migration dynamics to patient outcome are lacking. We investigated whether two computational frameworks identify a common prognostic dimension.

Methods: Single-cell trajectories from 27 primary IDH1-wild-type GBM cultures were monitored by time-lapse imaging. Data were analyzed using a physics-guided framework to extract population-level descriptors, including the mean squared displacement (MSD) scaling exponent α , defining the diffusion regime. Concurrently, a neural network autoencoder compressed 50-frame trajectory windows into a two-dimensional latent space, clustered (K-means, silhouette-validated) into three reproducible cell-level phenotypes. Per-line phenotype composition was correlated with α and overall survival using multivariate Cox regression and Kaplan–Meier analyses.

Results: Physics-guided analysis identified α as the only independent predictor of survival ($p=0.023$), with mortality risk increasing nonlinearly above $\alpha=1.04$. Independently, autoencoder-derived phenotype composition strongly correlated with α ($r=-0.56$, $p=0.004$): the low-motility "Confined" phenotype was associated with longer overall survival (log-rank $p=0.005$), whereas high-motility "Exploratory" phenotypes were associated with shorter survival ($p=0.025$).

Conclusions: Two independent approaches converged on the same biophysical axis: low-motility, subdiffusive dynamics marked a favorable clinical phenotype, whereas high-motility, superdiffusive dynamics identified a high-risk phenotype. These findings support cell-resolved dynamical profiling as a robust candidate prognostic biomarker in GBM.